Session 6: Computational Hydrodynamics in 1D

Euler equations, finite volume, the HLL flux, the Sod shock tube

astro-lab – reading notes

Status: reading notes and exercise specification. The code for this session (`06_hydro_1d/`) is not in the repository yet; the exercises below define what to build.

Goals

Write a finite-volume solver for the 1D Euler equations, validate it on the Sod shock tube against the exact solution, and measure how its error behaves as the grid is refined.

1 The Euler equations

Gas dynamics without viscosity, conduction or gravity is a system of conservation laws,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} = 0, \qquad \mathbf{U} = \begin{pmatrix} \rho \\ \rho u \\ E \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} \rho u \\ \rho u^2 + P \\ u(E+P) \end{pmatrix},$$

with total energy density $E = \dfrac{P}{\gamma - 1} + \frac{1}{2}\rho u^2$ for an ideal gas, and sound speed $c = \sqrt{\gamma P/\rho}$. Because the equations are in conservation form, mass, momentum and energy are conserved to round-off by any scheme that updates cells by flux differences. That is the main reason to use finite volume rather than finite differences on primitive variables.

2 Finite-volume update

Divide the domain into cells of width Δx with cell averages $\mathbf{U}_i$. Then

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_{i+1/2} - \mathbf{F}_{i-1/2}\right),$$

where $\mathbf{F}_{i+1/2}$ is a numerical flux computed from the states on the two sides of the interface. Stability requires the CFL condition

$$\Delta t = C\,\frac{\Delta x}{\max_i\left(|u_i| + c_i\right)}, \qquad C \le 1 \text{ (use 0.4–0.8).}$$

The HLL flux

Estimate the fastest left- and right-moving wave speeds at an interface (Davis estimates):

$$S_L = \min(u_L - c_L,\ u_R - c_R), \qquad S_R = \max(u_L + c_L,\ u_R + c_R).$$

Then

$$\mathbf{F}^{\mathrm{HLL}} = \begin{cases} \mathbf{F}_L & S_L \geq 0, \\ \dfrac{S_R\mathbf{F}_L - S_L\mathbf{F}_R + S_L S_R(\mathbf{U}_R - \mathbf{U}_L)}{S_R - S_L} & S_L < 0 < S_R, \\ \mathbf{F}_R & S_R \leq 0. \end{cases}$$

HLL is robust and simple but smears contact discontinuities because it ignores the middle wave. HLLC restores it. Lax–Friedrichs is the simplest flux and the most diffusive; it is a good first implementation to get the bookkeeping right.

First-order versus second-order. Using the cell values directly on both sides of an interface gives a first-order scheme: stable but diffusive. Reconstructing a linear profile in each cell with a slope limiter (minmod, MC) and a half-step predictor gives second order in smooth regions without spurious oscillations at shocks.

3 The Sod shock tube

Initial conditions on $x \in [0, 1]$, diaphragm at $x = 0.5$, $\gamma = 1.4$:

$$(\rho, u, P) = (1,\ 0,\ 1) \quad (x < 0.5), \qquad (\rho, u, P) = (0.125,\ 0,\ 0.1) \quad (x > 0.5).$$

At $t = 0.2$ the exact solution has a left-moving rarefaction fan, a contact discontinuity and a right-moving shock. Standard values for the intermediate states:

Quantity	Value
P_*	0.30313
u_*	0.92745
ρ_{*L} (left of contact)	0.42632
ρ_{*R} (right of contact)	0.26557

At $t = 0.2$ the features sit at approximately $x \approx 0.263$ (rarefaction head), 0.486 (rarefaction tail), 0.686 (contact) and 0.850 (shock).

4 Implementation plan

1. Arrays of conserved variables with two ghost cells; transmissive boundaries.
2. Functions `prim_to_cons`, `cons_to_prim`, `flux`, `hll_flux`.
3. Time loop with the CFL timestep, stopping exactly at $t = 0.2$.
4. Check: total mass, momentum and energy are conserved to round-off until waves reach the boundary.
5. Compare density, velocity and pressure with the exact solution (write a Riemann solver or use a published reference solution).
6. Convergence study: $N = 100, 200, 400, 800, 1600$; plot the L_1 density error versus Δx on log–log axes.

5 Exercises

1. Implement Lax–Friedrichs first and compare against HLL; describe the difference at the contact.
2. Add MUSCL reconstruction with a minmod limiter and measure the improvement.
3. Explain why the measured L_1 convergence rate at a solution with discontinuities is lower than the scheme's formal order.
4. Test a case with a strong shock-tube such as the Lax problem or a double rarefaction (123 problem) and note any failures (negative pressure) and how to guard against them.
5. Connect to Session 3: add a static gravitational source term and check that an isothermal atmosphere stays in hydrostatic balance.

References

- Toro, *Riemann Solvers and Numerical Methods for Fluid Dynamics*, ch. 4, 10
- LeVeque, *Finite Volume Methods for Hyperbolic Problems*
- Zingale, *Computational Hydrodynamics for Astrophysics* (lecture notes; search for the title)
- Sod (1978), *J. Comput. Phys.* 27, 1